MRTT Causal Analysis

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2026-04-16

Contents

1 Data Loading

```
## Episode data: 6800 rows, 34 seeds, 100 episodes/seed/world
```

```
## Round data: 1360000 rows from 68 runs
```

2 Analysis 1: Episode-Level G-Computation

Table 1: Episode-level point estimates

Outcome	Naive	G-comp A (Full)	G-comp A (Lean)	G-comp B
investor_return	-4018.53	-3811.21	-4202.61	-4018.53
fair_1_repay	0.00	-0.01	0.00	0.00
max_1_repay	0.00	0.01	0.00	0.00

```
##   ep boot [250/1000]
##   ep boot [500/1000]
##   ep boot [750/1000]
##   ep boot [1000/1000]
```

Table 2: Episode-level: Bias, RMSE, 95% CI Coverage

Outcome	Estimator	True ATE	Bias	RMSE	Coverage	SE

Investor Return	naive	-4018.5288	**-107.2046**	3176.9729	0.954	3176.7524
Investor Return	A_full	-4018.5288	**1819.5630**	12920.4363	0.945	12798.0723
Investor Return	A_lean	-4018.5288	**-354.0792**	3312.5542	0.947	3295.2240
Investor Return	B	-4018.5288	**-107.2046**	3176.9729	0.954	3176.7524
Fair Repay	naive	-0.0030	**0.0000**	0.0045	0.953	0.0045
Fair Repay	A_full	-0.0030	**-0.0052**	0.0239	0.954	0.0233
Fair Repay	A_lean	-0.0030	**-0.0003**	0.0047	0.949	0.0047
Fair Repay	B	-0.0030	**0.0000**	0.0045	0.953	0.0045
Max Repay	naive	0.0022	**-0.0001**	0.0046	0.951	0.0046
Max Repay	A_full	0.0022	**0.0107**	0.0242	0.945	0.0218
Max Repay	A_lean	0.0022	**-0.0008**	0.0052	0.941	0.0051
Max Repay	B	0.0022	**-0.0001**	0.0046	0.951	0.0046

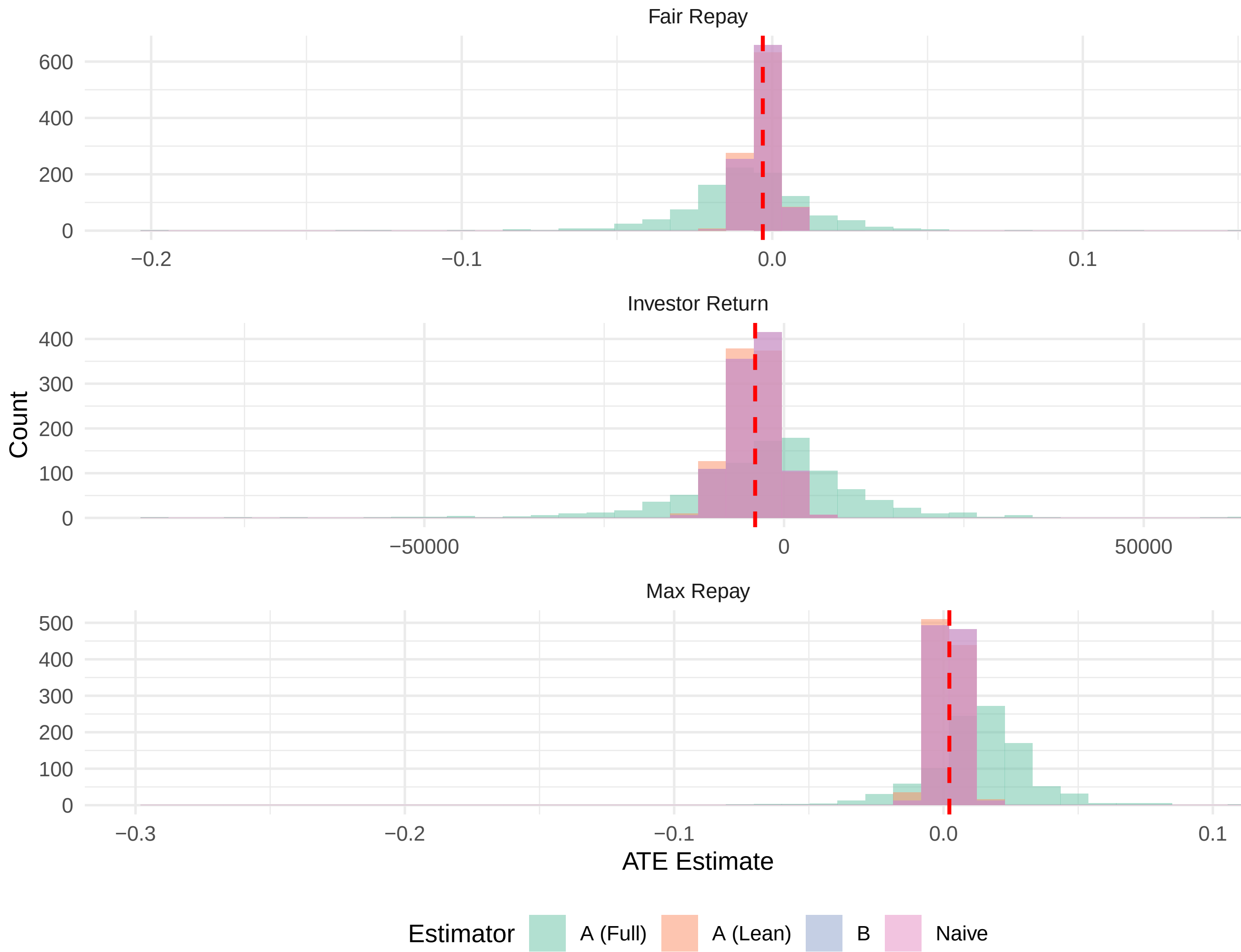

Figure 1: Episode-level bootstrap distributions

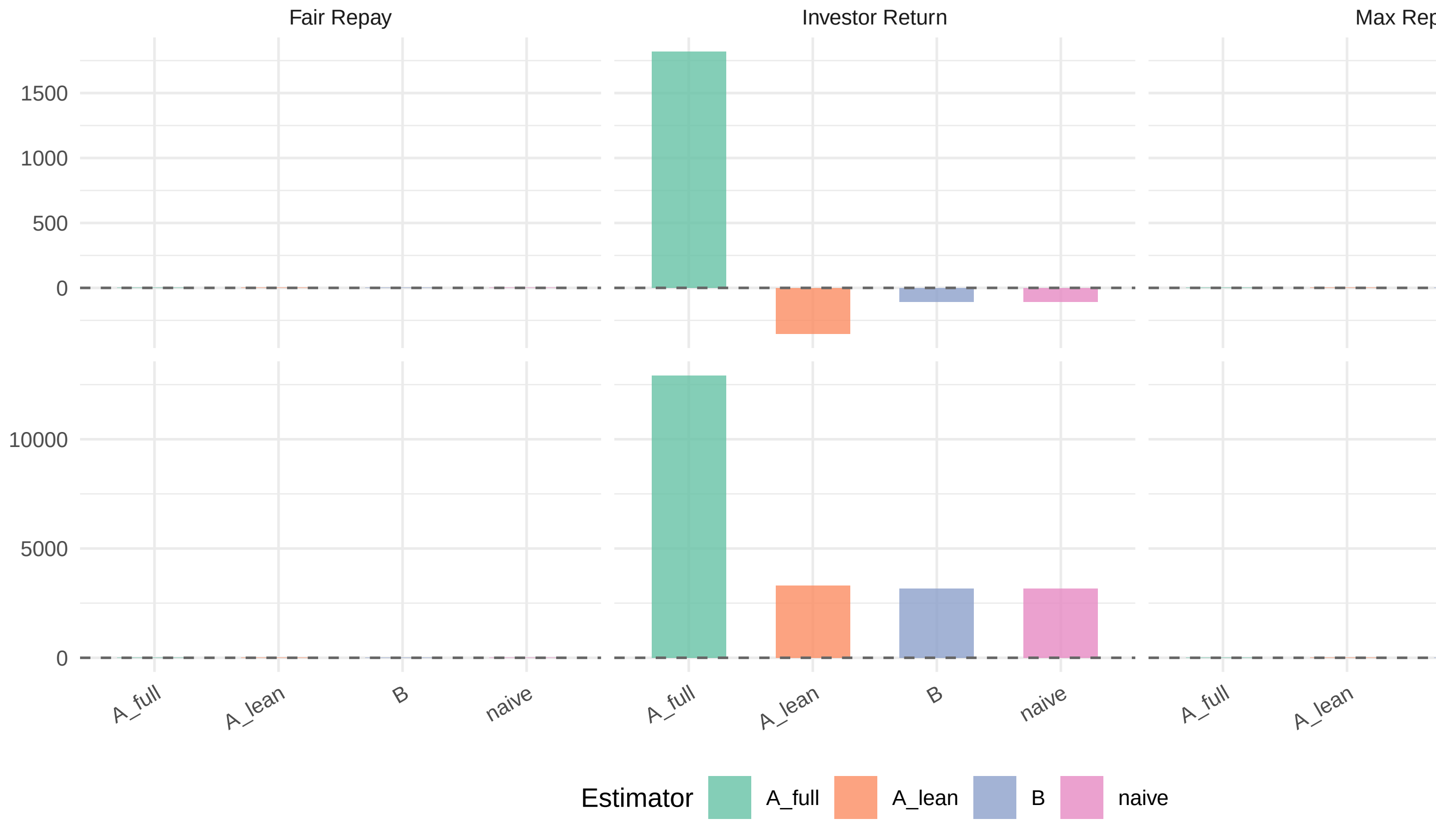

Figure 2: Episode-level bias and RMSE

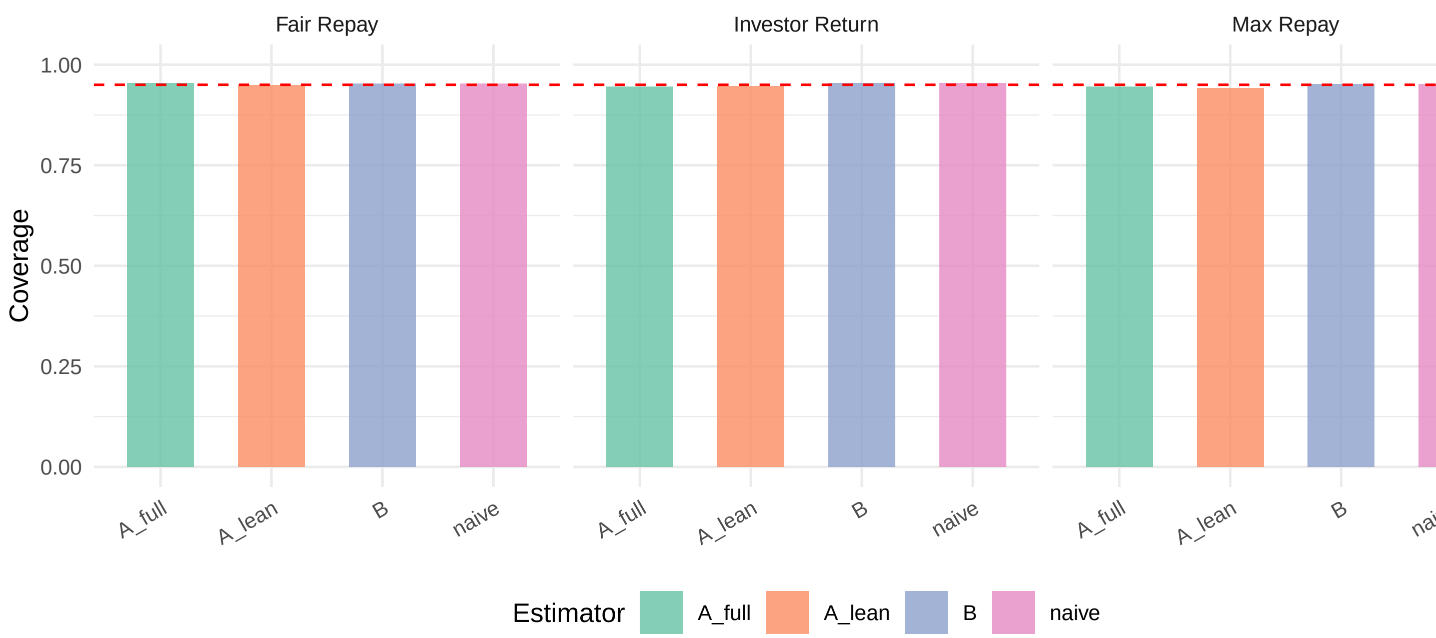

Figure 3: Episode-level 95% CI coverage

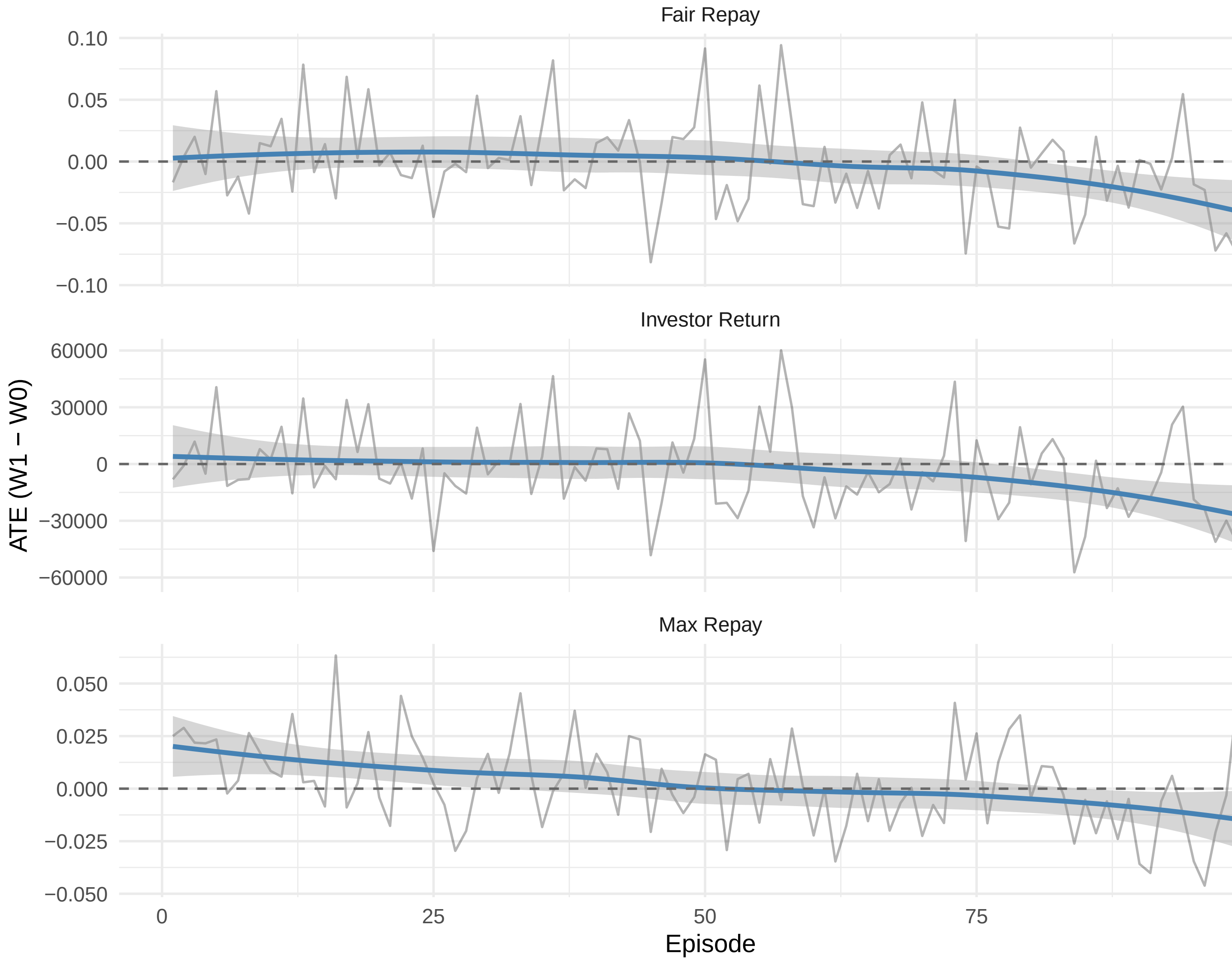

Figure 4: ATE by training episode

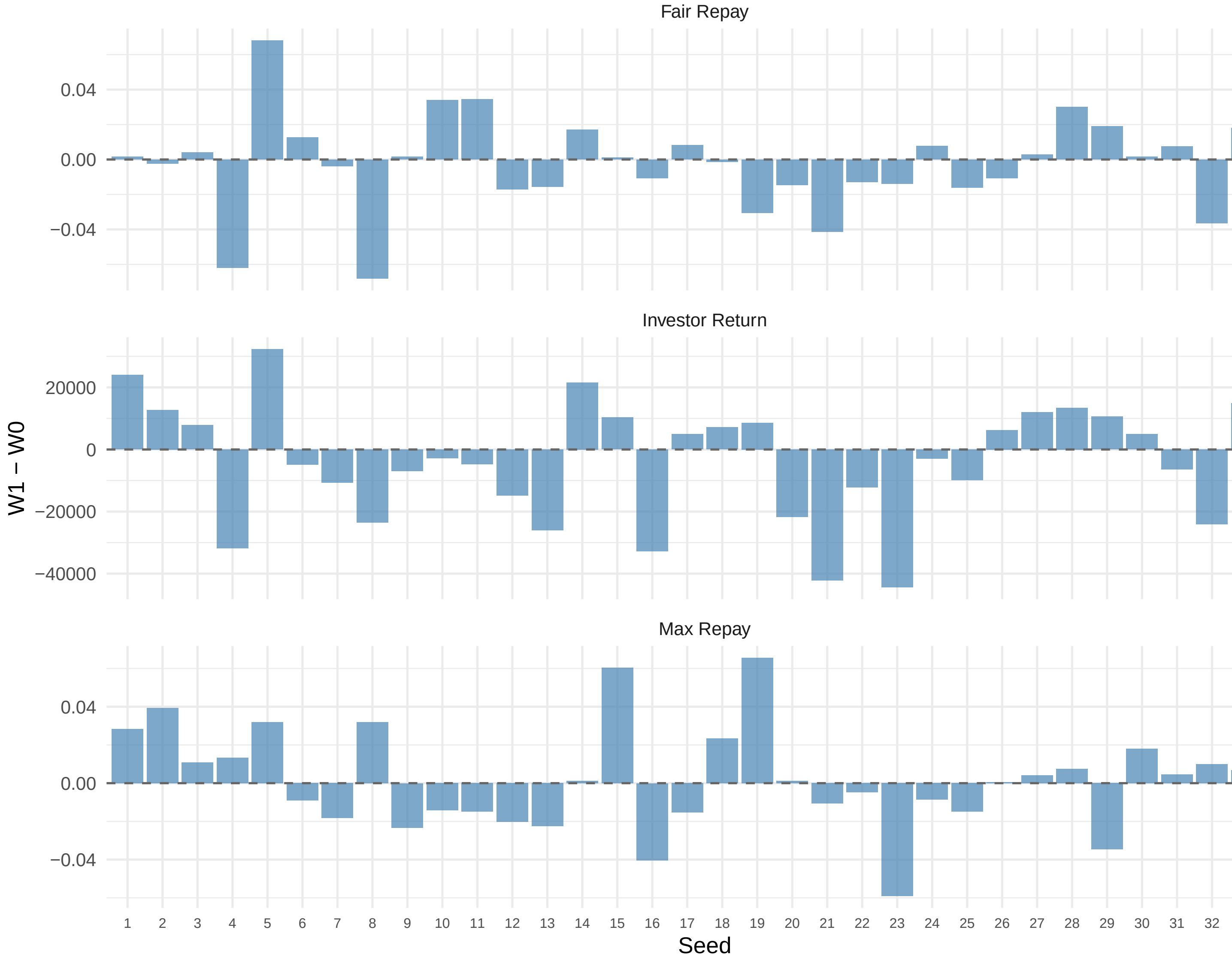

Figure 5: Within-seed paired differences

3 Analysis 2: Round-Level G-Computation

```
## Round data (subsampled): 27200 rows, h_cols found: 5
```

Table 3: Round-level ground truth ATEs

Outcome	True ATE
repayment_pct	-0.0013
investor_reward	0.4006

```
##   rd boot [250/1000]
##   rd boot [500/1000]
##   rd boot [750/1000]
##   rd boot [1000/1000]
```

Table 4: Round-level: Bias, RMSE, 95% CI Coverage

Outcome	Estimator	True ATE	Bias	RMSE	Coverage	SE
Repayment %	naive	-0.0013	**0.0000**	0.0226	0.951	0.0226
Repayment %	A	-0.0013	**0.0175**	0.0226	0.792	0.0144
Repayment %	B	-0.0013	**0.0079**	0.0200	0.931	0.0184
Investor Reward	naive	0.4006	**0.0136**	1.2443	0.942	1.2448
Investor Reward	A	0.4006	**0.9222**	1.2561	0.822	0.8532
Investor Reward	B	0.4006	**0.6529**	1.1762	0.905	0.9789

4 Combined Summary

Table 5: Combined evaluation across both analysis levels

Level	Outcome	Estimator	True ATE	Bias	RMSE	Coverage	SE
Episode	Investor Return	ep_naive	-4018.5288	**-107.2046**	3176.9729	0.954	3176.7524
Episode	Investor Return	ep_A_full	-4018.5288	**1819.5630**	12920.4363	0.945	12798.0723
Episode	Investor Return	ep_A_lean	-4018.5288	**-354.0792**	3312.5542	0.947	3295.2240
Episode	Investor Return	ep_B	-4018.5288	**-107.2046**	3176.9729	0.954	3176.7524
Episode	Fair Repay	ep_naive	-0.0030	**0.0000**	0.0045	0.953	0.0045
Episode	Fair Repay	ep_A_full	-0.0030	**-0.0052**	0.0239	0.954	0.0233
Episode	Fair Repay	ep_A_lean	-0.0030	**-0.0003**	0.0047	0.949	0.0047
Episode	Fair Repay	ep_B	-0.0030	**0.0000**	0.0045	0.953	0.0045
Episode	Max Repay	ep_naive	0.0022	**-0.0001**	0.0046	0.951	0.0046
Episode	Max Repay	ep_A_full	0.0022	**0.0107**	0.0242	0.945	0.0218
Episode	Max Repay	ep_A_lean	0.0022	**-0.0008**	0.0052	0.941	0.0051
Episode	Max Repay	ep_B	0.0022	**-0.0001**	0.0046	0.951	0.0046
Round	Repayment %	rd_naive	-0.0013	**0.0000**	0.0226	0.951	0.0226
Round	Repayment %	rd_A	-0.0013	**0.0175**	0.0226	0.792	0.0144
Round	Repayment %	rd_B	-0.0013	**0.0079**	0.0200	0.931	0.0184
Round	Investor Reward	rd_naive	0.4006	**0.0136**	1.2443	0.942	1.2448

Round	Investor Reward	rd_A	0.4006	**0.9222**	1.2561	0.822	0.8532
Round	Investor Reward	rd_B	0.4006	**0.6529**	1.1762	0.905	0.9789

Figure 6: Round-level bootstrap distributions

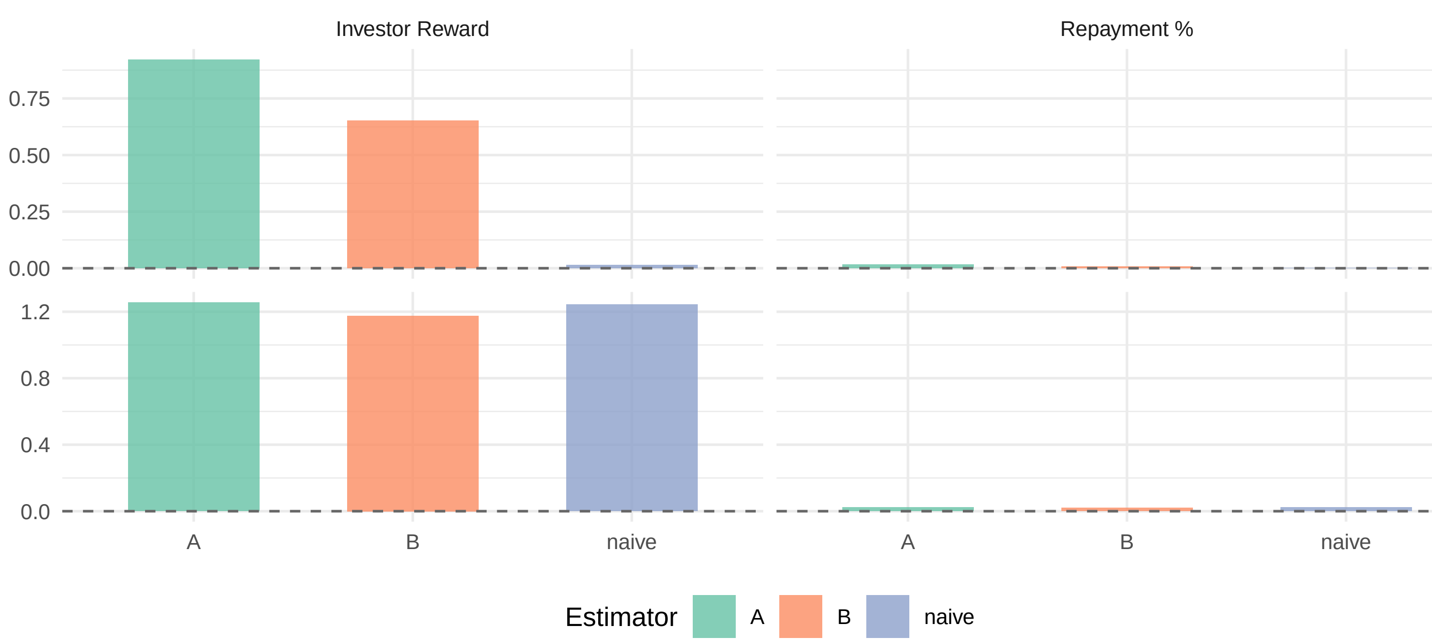

Figure 7: Round-level bias and RMSE